DATA 201
Data Communication and Ethics

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PARAMI UNIVERSITY

Instructor Profile

Ph.D. in Computer Science with expertise in **natural language processing**

Research focus on **responsible AI and natural language processing**

Experience applying data science methods to **real-world social and healthcare problems**

Teaching philosophy emphasizes **clarity, integrity, and ethical responsibility in data-driven decision making**

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Class Schedule

Monday - 7:00 - 8:30 PM (Thailand Time)

Wednesday - 7:00 - 8:30 PM (Thailand Time)

Learning Outcome

CLO1. Apply statistical reasoning to explore, summarize, and interpret data, accounting for uncertainty and variability.

CLO2. Select and justify appropriate statistical methods and data visualizations based on data characteristics and analytical objectives.

CLO3. Communicate statistical findings clearly using precise language, visual evidence, and quantitative summaries for both technical and non-technical audiences.

CLO4. Critically evaluate data analyses and visualizations for correctness, bias, misinterpretation, and misleading representations.

CLO5. Demonstrate ethical data science practices by addressing issues of data privacy, fairness, transparency, and responsible use of statistical results.

Learning Modules

Module 1: Statistical Reasoning & Exploratory Analysis (Weeks 1–4)
Module 2: Data Visualization & Communication (Weeks 5–8)
Module 3: Ethics, Bias, and Responsible Data Science (Weeks 9–12)

Evaluation

Weekly Activities/Assignments/Lab - 30%
Mid Term Mini Project - 20%
Final Project (Report+Presentation) - 30%
Participation & Peer Review - 20%

Code of Conduct

Academic Integrity: Submit original work, properly cite data, code, and sources, and disclose any permitted use of AI tools.

Ethical Data Use: Use data responsibly, respecting privacy, consent, fairness, and legal or institutional guidelines.

Honest Communication: Present statistical results and visualizations accurately, clearly stating assumptions, uncertainty, and limitations.

Respectful Environment: Engage respectfully with instructors and peers; discrimination, harassment, or disruptive behavior is not tolerated.

Professional Conduct: Collaborate fairly, meet deadlines, provide constructive peer feedback, and follow course policies and instructions.

Polling Question

How do you understand data communication?

A. Presenting data using charts, tables, and dashboards

B. Explaining statistical results in words

C. Telling a story that connects data, analysis, and decision-making

D. Translating complex data insights for different audiences

E. I’m not sure yet

Data Communication

- **Data communication** is the process of **transforming data and statistical results into clear, accurate, and meaningful messages** that enable understanding, interpretation, and decision-making by a specific audience.

- In statistics and data science, data communication is **not just showing results**, but **explaining what the results mean, how reliable they are, and how they should be used**.

Why Data Communication Matters

- Data analysis has **no impact** if it is not understood
- Poor communication can lead to:
  - Wrong decisions
  - Misinterpretation of statistical results
  - Loss of trust in data and analysts
- Statisticians and data scientists are **responsible interpreters**, not just number producers

Issue with data communication

Context: Media coverage of AI and data science research.

Typical headline

> "AI system achieves 99% accuracy"

Missing information ⇒ Dataset size and balance, Definition of "accuracy", Performance on minority or rare cases
Why this is dangerous ⇒ Overconfidence in model deployment, Public misunderstanding of AI reliability

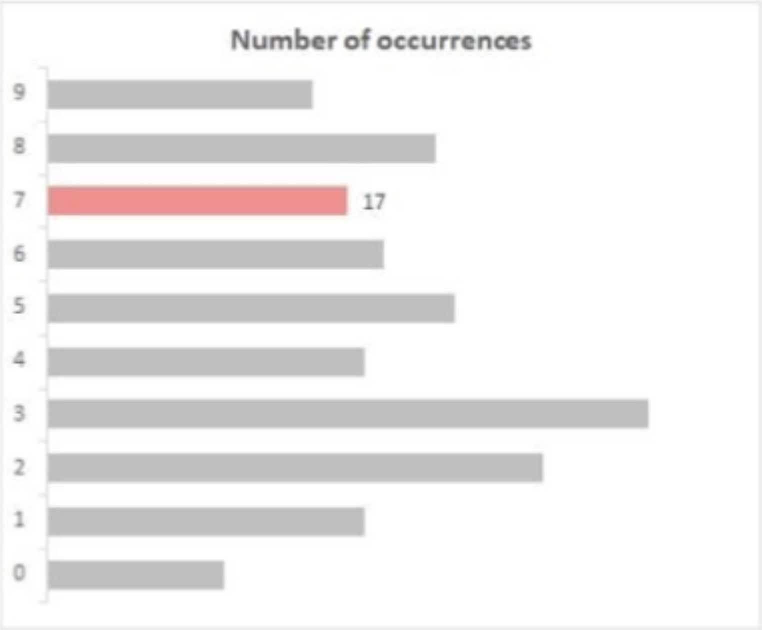
Responsible data communication ⇒ Limitations, Failure cases, Appropriate use scenarios

Data visualization example

Count how many times the number 7 appears in the grid below.

5	2	8	3	6	1	9	3	6	2	5	3	7	4	3	8	3
8	5	8	9	6	2	1	4	4	3	9	3	6	5	2	4	9
1	0	2	7	5	2	8	3	6	1	6	2	9	3	8	3	8
5	8	4	7	2	0	3	7	3	5	4	7	1	8	2	0	1
2	5	3	6	4	3	9	1	0	8	9	5	7	3	4	5	3
2	7	5	2	8	3	6	1	6	2	9	3	8	3	8	5	8
4	7	2	0	3	7	3	5	4	7	1	8	2	0	1	9	6
2	1	4	4	3	9	3	6	5	2	4	9	1	0	2	7	5
2	8	3	6	1	6	2	9	3	8	3	8	5	8	4	7	2
0	3	7	3	5	4	7	1	8	2	0	1	2	5	3	6	4
3	9	1	8	9	5	0	7	3	4	5	3	2	7	5	2	8
3	6	1	6	2	4	6	2	7	5	9	1	5	2	6	3	6

Data visualization example



5	2	8	3	6	1	9	3	6	2	5	3	7	4	3	8	3
8	5	8	9	6	2	1	4	4	3	9	3	6	5	2	4	9
1	0	2	7	5	2	8	3	6	1	6	2	9	3	8	3	8
5	8	4	7	2	0	3	7	3	5	4	7	1	8	2	0	1
2	5	3	6	4	3	9	1	0	8	9	5	7	3	4	5	3
2	7	5	2	8	3	6	1	6	2	9	3	8	3	8	5	8
4	7	2	0	3	7	3	5	4	7	1	8	2	0	1	9	6
2	1	4	4	3	9	3	6	5	2	4	9	1	0	2	7	5
2	8	3	6	1	6	2	9	3	8	3	8	5	8	4	7	2
0	3	7	3	5	4	7	1	8	2	0	1	2	5	3	6	4
3	9	1	8	9	5	0	7	3	4	5	3	2	7	5	2	8
3	6	1	6	2	4	6	2	7	5	9	1	5	2	6	3	6

Exploratory data analysis (EDA)

•Exploratory data analysis (EDA) is the process of reviewing new data to discover patterns, to spot anomalies, to test hypotheses, and to check assumptions.

•Just like in our counting example, EDA is made easier by using summary statistics and data visualizations.•Often, summary statistics alone aren’t enough.

EDA Example

MONTH	SALES REP 1	SALES REP 2	SALES REP 3
1-Jan	101	34	171
2-Feb	199	111	354
3-Mar	300	633	460
4-Apr	405	313	370
5-May	495	409	145
SUM	1,500	1,500	1,500
AVERAGE	300	300	300

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Understanding Data

What is data?

Definition of *data*

1 : factual information (such as measurements or statistics) used as a basis for reasoning, discussion, or calculation

// the ***data*** *is plentiful and easily available*

— H. A. Gleason, Jr.

// comprehensive ***data*** *on economic growth have been published*

— N. H. Jacoby

2 : information in digital form that can be transmitted or processed

3 : information output by a sensing device or organ that includes both useful and irrelevant or redundant information and must be processed to be meaningful

Data Types

- Qualitative
  - Descriptive, Non-numeric
  - Examples: color, category, texture
  - May be subjective
  - Nominal
    - Categories with no order (e.g., gender, race, yes/no)
    - Can calculate frequency, not numerical operations
  - Ordinal
    - Ranked categories (e,g., education level, staff position)
- Quantitative
  - Numeric, Measurables
  - Examples: distance, time, price, temperature
  - Discrete
    - Countable whole numbers (e,g., number of products sold)
  - Continuous
    - Measurable with infinite values (e.g., time, distance)

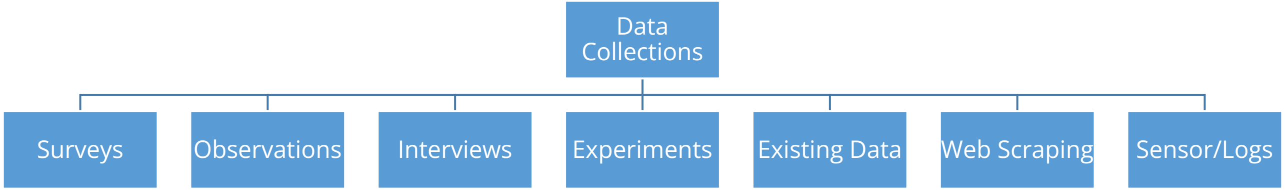
Data Sources

- Primary Data
  - Collected firsthand
  - Surveys, experiments, app data
- Secondary Data
  - Collected by others
  - Government reports, third-party organizations data etc.

Data Collection Methods

What is Data Collection?

The systematic process of gathering and measuring information on targeted variables



Data Collection Methods

- **Surveys:** Questionnaires (online, paper, interviews).
  - *Pros:* Collect data from large samples.
  - *Cons:* Potential for biased responses, low response rates.
- **Observations:** Watching and recording behavior or events.
  - *Pros:* Can provide direct, unfiltered information.
  - *Cons:* Can be time-consuming, observer bias.
- **Interviews:** One-on-one or group discussions.
  - *Pros:* In-depth understanding, flexible.
  - *Cons:* Time-intensive, interviewer influence.
- **Experiments:** Controlled studies to determine cause-and-effect.
  - *Pros:* High internal validity.
  - *Cons:* Can be artificial, ethical considerations.

Data Collection Methods

- **Existing Data (Secondary Data):** Using data already collected by others.
    - *Pros:* Cost-effective, time-saving.
    - *Cons:* Data might not perfectly fit needs, quality concerns.
- **Web Scraping:** Extracting data from websites.
    - *Pros:* Access to vast amounts of public data.
    - *Cons:* Technical complexity, ethical/legal issues.
- **Sensors & Logs:** Automated data collection (e.g., IoT devices, server logs).
    - *Pros:* Real-time data, high volume.
    - *Cons:* Requires infrastructure, potential for noise.

Data Collection Ethics

- Informed Consent
- Confidentiality
- Data Ethics Law
  - GDPR (EU)
  - CCPA (US)
  - PDPA (Thailand)

Why Clean Data?

"Garbage In, Garbage Out" (GIGO)

- **Accuracy:** Ensures data reflects reality.
- **Consistency:** Uniformity in data formats and values.
- **Completeness:** Addresses missing information.
- **Reliability:** Trustworthy data for decision-making.
- **Validity:** Data conforms to defined rules and constraints.
- **The Goal:** Prepare data for effective analysis.

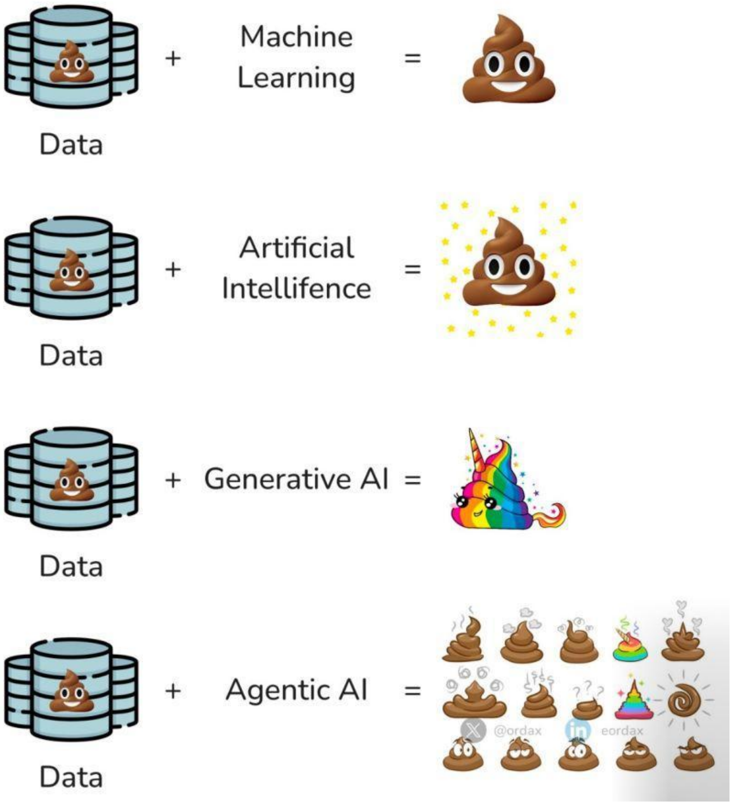


Image Credit: Eduardo Ordax

Handling Missing Data

- Common reasons of missing data
    - No response in survey forms
    - Data is not available
    - Data entry errors (spelling mistakes)
    - Irrelevant data

Deletion

- Listwise/Casewise Deletion: Remove entire rows with any missing values.
  - *Caution:* Can lead to significant data loss and biased results if data is not missing completely at random (MCAR).
- Pairwise Deletion: Use available data for specific analyses (e.g., correlation between two variables).
- Column Deletion: Remove entire columns if most values are missing.

Deletion

UserID	Age	Feature Usage (Hours/Week)	Satisfaction (1-5)	Subscription	Feedback Submitted (YES/NO)
U001	25	5	4	Premium	YES
U002	32	Null	5	Null	NO
U003	Null	2	3	Basic	YES
U004	45	10	Null	Null	YES
U005	28	7	4	Null	YES
U006	30	3	2	Null	NO
U007	22	1	1	Null	YES
U008	50	8	5	Premium	NO

Deletion

- Listwise/Casewise
  - U002, U003, U004, U005 (each have at least one missing value) will be removed
- Pairwsie: Correlation between Age and Feature Usage
  - U002: Feature usage is Null (Do not use for analysis)
- Column Deletion
  - Subscription tier column has 5 out of 8 values missing (62.5% missing)
  - If a threshold is set (e.g., delete column if more than 50% missing data), the Subscription will be deleted.

Data Imputation

CustomerID	Age	Email	LastPurchaseDate	Region
1	34	alice@email.com	2024-05-15	North
2	Null	bob@email.com	2024-03-20	South
3	45	Null	2024-06-01	East
4	28	charlie@email.com	Null	North
5	52	david@email.com	2024-04-10	Null
6	34	eve@email.com	2024-05-25	West

Data Imputation

- Mean/Median Imputation for Age
  - Mean = 38.6 (Replace Null in Age for Customer ID2 with 38.6)
  - Median = 34 (28, 34, 34, 45, 52) – Replace Null in Age for CustomerID 2 with 34
- Mode Imputation for Region
  - North was the most frequent region, missing Region for Customer ID 5 could be filled with 'North'.

Mean OR Median

- You're analyzing the monthly incomes (in USD) of a small group of 7 people, but one person didn’t report their income:
- Incomes = [2500, 2600, 2700, None, 2800, 3000, 50000]
- Mean = 10600
- Median = 2750 (2700+2800/2) - **Observation**: Median is **not affected** by the extreme value (50000), so the imputed value is **more realistic**.

Method	Sensitive to Outliers?	Good for…	Risk
Mean Imputation	Yes	Symmetric (normal) data	Skewed values distort mean
Median Imputation	No	Skewed data or outliers	Ignores small variations

Data Transformation

- Data transformation is the process of converting data into a format that is more appropriate or useful for analysis or machine learning.
- **Why it matters**:
  - Machine learning models often expect numeric, scaled, or encoded inputs
  - Helps improve model performance
  - Makes data easier to visualize and interpret

Data Transformation

Product	Price	Rating	Sales	Category
A	100	3.5	1000	Electronics
B	250	4.0	1500	Electronics
C	500	4.5	2000	Clothing
D	1000	3.0	3000	Clothing
E	2000	5.0	10000	Clothing

Min-Max Normalization

- Normalize into a specified range (Price)

Formula:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Where:

- $x_{\min} = 100$
- $x_{\max} = 2000$

Example: For Product C (Price = 500)

$$x_{\text{norm}} = \frac{500 - 100}{2000 - 100} = \frac{400}{1900} \approx 0.2105$$

Product	Price	Price (Normalized)
A	100	0.0000
B	250	0.0789
C	500	0.2105
D	1000	0.4737
E	2000	1.0000

Standardization Z Score

- Standardizing allows comparing how far a score is from the subject average.
- Z-score for Rating

Formula:

$$z = \frac{x - \mu}{\sigma}$$

Where:

- $\mu = \text{mean} = \frac{3.5+4.0+4.5+3.0+5.0}{5} = 4.0$
- $\sigma = \text{std dev} = \sqrt{\frac{(3.5-4)^2+(4-4)^2+(4.5-4)^2+(3.0-4)^2+(5.0-4)^2}{5}} \approx 0.7906$

Example: For Product D (Rating = 3.0)

$$z = \frac{3.0 - 4.0}{0.7906} \approx -1.265$$

Product	Rating	Rating (Z
A	3.5	-0.6325
B	4.0	0.0000
C	4.5	+0.6325
D	3.0	-1.2650
E	5.0	+1.2650

Getting Started

How to communicate using data?

- Know your audience and understand how it processes visual information. (Who)

- Determine what you’re trying to visualize and what kind of information you want to communicate. (What)

- Choose a type of visual that conveys the information in the best and simplest form for your audience. (How)

Know Your Audience (KYA)

- Know your audience and understand how it processes visual information.
- Consider audience familiarity. For example:
  - *High-level executives are generally well-versed in visual data, so use a variety of methods to stand out*
  - *Less-experienced audiences will want it kept simple (e.g., pie charts, bar graphs, and word maps)*
- –Consider how the visualization will be used by the audience:
  - –*Is it for executives to use to make decisions?*
  - –*Is it to inform the public?*

Activity

1. https://www.theguardian.com/media/2026/jan/31/can-you-guess-our-screen-time-a-priest-pensioner-tech-ceo-and-teenager-reveal-all
2. https://www.ft.com/content/faeb72a0-62ce-4d60-968e-5b932ea9c7e4
3. https://www.bbc.com/news/articles/cn40j0772vwo
4. https://www.foxnews.com/health/deadly-cancer-risk-spikes-certain-level-alcohol-consumption-study-finds

Discussion

What data is used? - Counts, percentages, averages, trends, comparisons?

What is the intended takeaway? - What conclusion does the article suggest?

What is missing or unclear? - Sample size? Time period? Population? Uncertainty or variability? Definitions?

Is the communication potentially misleading? Overgeneralization? Confusing correlation with causation? Emotional framing? Misleading visuals or headlines?

Who is the audience? Is the explanation appropriate for that audience?

Thank You

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